

From Leiden to Pleasure Island: The Constant Potts Model for Community Detection as a Hedonic Game

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November 7, 2025

Clusters or Community

- Unsupervised Machine Learning Technique
- Networks
- Non-Overlapping Partitions
- Competing Objectives
 - Maximize Friends
 - Minimize Strangers
- Frustrated Choice (Cannot Max/Min both)

- A node is **robust** if its current community is unambiguously preferred over all other communities.
- For a given partition, the robustness will be the fraction of nodes that there is no better move for.

Familiarity Index

$$F_i(C_A, C_B) = \frac{\Delta d}{\Delta d + \Delta \hat{d}}$$

- $F > 1$ or $F < 0$ you should make that change
- $0 < F < 1$ are frustrated choices

Constant Potts Model

Comes from an analog of the Hamiltonian for the magnetic spin in statistical mechanics.

$$v_{ij} = (A_{ij} - \gamma) \delta_{v_i, v_j \in C_k}$$

Can vary γ :

- $\gamma = 0$ ignores strangers
- $\gamma = 1$ ignores friends
- Frustrated nodes
 - $\gamma < F_i$ go to more friends
 - $\gamma > F_i$ go to less strangers

- Agents (nodes here) act selfishly
- Cooperation happens (stable communities emerge)
- Nash Equilibrium happens
- Equilibrium is beneficial for collective

- Hedonic Potential created
- Utility Functions can be computed in polynomial time
- Three scales:

① Node

$$\varphi_i^\gamma(C_k) = \sum_{j \in C_k} v_{ij}$$

② Community

$$\phi^\gamma(C_k) = \frac{1}{2} \sum_{i \in C_k} \sum_{j \in C_k} v_{ij}$$

③ Partition

$$\Phi^\gamma(\pi) = \sum_{k=1}^K \phi^\gamma(C_k) = (1 - \gamma) \sum_{k=1}^K m_k - \gamma \sum_{k=1}^K \left(\binom{n_k}{2} - m_k \right)$$